User-Centered Microtransit for Adolescents

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***Abstract*—This study investigates a user-centered microtransit service specifically for transporting adolescents to after-school activities. The methodology includes generating a synthetic student population and the corresponding trip requests and solving a Vehicle Routing Problem with Pickup and Delivery with Time Windows (VRPPDTW) for route generation, considering students 'schedules and drivers' mandatory breaks. The case study focuses on the WeeDrive project, which will introduce an on-demand transport service in Limassol, Cyprus to help families manage their teenagers' after-school schedules. This service will operate with pre-booking and a dedicated fleet of private minibuses. Various scenarios are analyzed to assess the impact of different service strategies on the number of students served while ensuring that all trip requests are accepted, as well as the economic viability of the service with a particular focus on user satisfaction.**

***Index Terms*—on-demand transport, microtransit, VRPPDTW**

I. Introduction

Cities today are prioritizing the reduction of traffic congestion and emissions by promoting more sustainable transport options, such as shared and active modes of transport. Encouraging a shift away from private vehicles involves strategies like improving public transit and creating pedestrian- and cyclist-friendly environments. However, specific groups may resist this transition; for instance, parents often have safety concerns about unaccompanied travel for underage students. In this context, on-demand transport services offer a promising solution, providing a safe, flexible, and efficient alternative to private cars [1]. Microtransit is an on-demand transport service that uses minibuses or similar vehicles and typically operates in predefined areas as a first-/last-mile connection to public transport hubs or as a complementary service to fixed-route transit, either periodically or during non-recurrent events such as train malfunctions or road traffic incidents. Microtransit can also offer door-to-door service, allowing for improved convenience and user satisfaction. Microtransit in practical applications has to overcome several hurdles with respect to the financial sustainability, given the fact that the satisfaction of users and operators depend on competing performance metrics, where high utilization of vehicles might lead to decreased user satisfaction due to the additional travel times associated with the multiple detours necessary to maintain high cost efficiency [2]. Vehicle automation is expected to alleviate these hurdles by decreasing operation cost and removing driver constraints such as working hours and mandatory breaks [3], however, automation has encountered delays and alternative ways to balance user and operator satisfaction should be considered.

This study investigates a user-centered microtransit service specifically for transporting students to after-school activities in the city of Limassol, Cyprus. The case study focuses on intra-urban trips to after-school activities, in contrast to the study in [4], which focused on trips to rural areas from home to school during the morning period. This is justified by the fact that in Limassol, the majority of after-school activities require arrival in time windows characterized by limited flexibility and the burden of transporting students lies, in most cases, on the shoulders of parents, which has an adverse effect on the parents' quality of life but most importantly traffic conditions. Activities are distributed over a period of 8 hours, in effect preponing the start and extending the end of the afternoon peak period for several disparate locations within the urban traffic network of Limassol, corresponding to popular after-school activity locations.

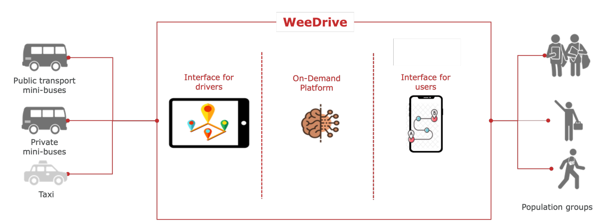


Fig. 1: Supply-Demand Matching Platform for on-Demand Shared Services

In this study, we evaluate the feasibility and economic viability of microtransit, initially targeting adolescent students in the city of Limassol. The WeeDrive platform will be introduced as part of the microtransit service in Limassol, Cyprus, initially to help families manage the after-school schedules of their adolescent children, but

TABLE I: Abbreviations

Abbreviation	Description
EXP_P	Expected pick-up
EXP_D	Expected drop-off
ACT_S	Start of activity
ACT_E	End of activity
TT	Expected travel time
MD	Maximum detour
MW	Maximum waiting time

TABLE II: Time windows definition.

Window Type	Location	Input	Time window
EXP_P	home	(ACT_S-TT)	[(EXP_P-MD),EXP_P]
	school/activity	ACT_E	[EXP_P,(EXP_P+MW)]
EXP_D	activity	ACT_S	[0, EXP_D]
	home	(ACT_E+TT)	[0, (EXP_D+MD)]

ultimately expanding to additional demographics, such as commuters and tourists, as presented in Figure 1. This service will provide shared on-demand door-to-door transport to students, with the goal of alleviating parents from the burden of transporting their offspring to their after-school activities. This service will operate with pre-booking and a dedicated fleet of private minibuses. Minibuses can offer improved flexibility within the urban landscape, while still being capable of maintaining a high utilization rate in an on-demand, door-to-door service. The methodology we use starts with the generation of a synthetic student population, which we combine with a travel behavior survey [5] and real-world data collected from open-source datasets to create the after-school activities demand model for adolescent students between 12 and 18 years of age. We subsequently frame the routing problem as a Vehicle Routing Problem with Pickup and Delivery with Time Windows (VRPPDTW), considering adolescent students' schedules and drivers' mandatory breaks. Various scenarios are analyzed to assess the impact of different fares and service strategies on the number of students served while ensuring that all trip requests are accepted, as well as the economic viability of the service.

II. Problem Description and Solution Approach

A. Methodology

Our problem focuses on a passenger service operated using a fixed fleet of minibuses. The driver associated with each minibus is entitled to a break, which must be taken at any location within a certain time window. Drivers follow a predefined route that starts and ends at the bus depot. To ensure service quality, specific time windows are defined for passenger pick-ups and drop-offs. These windows are defined based on activity schedules, expected travel times between the origins and destinations of the trip, and service parameters, that is, the maximum waiting time and the maximum detour time for picking up and dropping off other passengers (see notation and the corresponding equations in Table II).

The methodology comprises two stages. The first stage has the objective of generating trip requests. It starts with the creation of synthetic population based on the distribution of students per school and on the home addresses. Then, a daily plan for each student is defined consisting of at least two trips: one to get to the activity (school/home-activity) and another to return (activity-home). In addition to those, daily plans can also include a trip from school to home (school-home) and a trip for an intermediate activity (activity-activity). In the second stage, route generation for transporting the adolescents to their after-school activities is framed as a VRPPDTW. Inspired by [6] the objective function is augmented to heavily penalize unassigned trip requests and missed breaks, in adddition to minimizing total travel distance.

B. Solution

School Bus Routing problems framed as VRPPDTW can be solved with many different approaches, where each approach can be more or less suitable for certain use cases [7]. Analytical solutions for small to medium size use cases can be found by employing methods such as ILP (Integer Linear Programming) [8], however, for larger size, higher complexity use cases, heuristics and metaheuristic solution approaches are employed [9]. In our use case, the VRPPDTW is solved using JSPRIT, a Java-based optimization library that employs a metaheuristic based on the Ruin-and-Recreate principle [10]. This method is a large neighborhood search that combines simulated annealing and threshold-accepting techniques. It starts with an initial solution and follows a "ruin-and-recreate" process: first dismantling parts of the solution, creating unassigned jobs and a partial solution, and then reintegrating the unassigned jobs to form a new solution. If the new solution meets quality criteria, it becomes the best solution, and the process repeats until a termination condition, such as a time limit or iteration count, is met.

III. Experiments

A. Design

The methodology is applied to investigate an on-demand service for students (WeeDrive) that operates on school days between 13:00 and 21:00. A fleet of 10 Ford Transit minibuses with 17 passenger seats is available. Students can be picked up from their schools or homes, with public schools typically finishing at 13:30

and private schools around 15:00. According to a survey conducted in [5], more than 80% of after school activities for Cyprus students involve tutorial lessons or sports activities. Among these students, 55% participate in both tutorial lessons and sports activities, while 26% attend only tutorial lessons, and 19% are involved only in sports. For trip generation, it is assumed that students cannot combine two sport activities, and the maximum interval between the end of one activity and the start of the next is 1 hour. Furthermore, student trips to and from activities must fall within a distance range of 750m to 10km. The service characteristics vary according to the typology of the origin and destination. For trips from home, pick-ups can occur up to 60 minutes before the start of an activity while for trips from schools or other activities, waiting times are limited to 30 minutes after the previous school/activity ends. Trips returning home from activities are allowed an extended travel time of up to 60 minutes. The locations of schools, homes, activity facilities, and the bus depot are shown in Figure 2

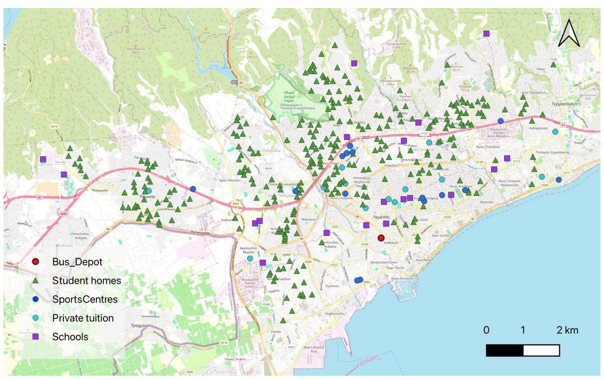


Fig. 2: Selected locations in Limassol, Cyprus

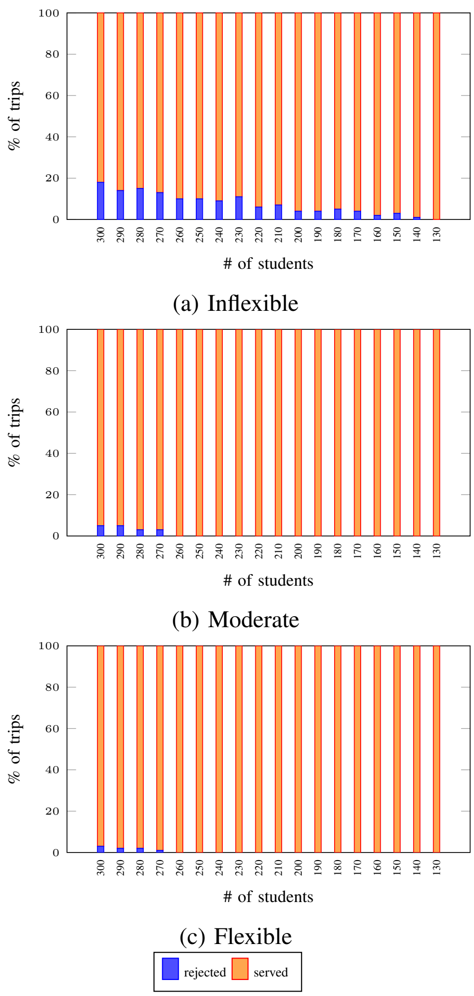
Operational costs estimates include a daily platform cost and fleet operator costs of €4200 including tax. Different service strategies are analyzed with respect to strictness levels in the pick-up and drop-off time windows and driver's schedules times. Three strictness levels are defined: inflexible (30-minute limits for both waiting times and detours), moderate (30-minute limit for waiting times and 60-minute limit for detours), and flexible (60-minute limits for both). Drivers start either simultaneously at 13:00 or staggered between 13:00 and 15:00. In compliance with EU regulations, drivers must take a break within 4.5 hours of driving, which can be either a single 45-minute period or split into 15-minute and 30-minute segments.

The pricing structure consists of per trip charges and is calculated ex post the VRPDTW solution, based on 3 distance-based pricing policies. In order to preserve fairness and equity in our pricing, the distance D in the pricing policies is not set as the actual distance $\hat{D}$ traveled during each trip, but as the driving distance from each trip origin to destination, without any detours. In the first and second policies, the fare per trip consists of an initial fee plus a constant distance-based fare. The first policy (Pol_1) uses a smaller initial fee of €4.50, with a higher distance-based increment ($\geq$ €0.50), meaning students who travel longer distances are more penalized. In contrast, the second policy (Pol_2) uses a higher initial fare of €6 and a smaller distance-based fare ($<$ €0.50), ensuring a more balanced fare for all students. In the third policy (Pol_3), the pricing includes not only the initial fee and distance-based fare but also the average occupancy per vehicle ($AOpv$). Specifically, unless the vehicle carries only one passenger, there is a partial overlap of trips. An equitable way to distribute cost savings is to divide the initial fare by the average occupancy per vehicle ($AOpv$). Therefore, due to this division, this policy is associated with the highest initial fare of €16.6. All trip costs in these policies are capped at local taxi fare rates. The rationale behind this upper bound is that parents are unlikely to pay more than the equivalent taxi fare, given the higher level of comfort taxis offer compared to minibuses. The distance-based fares are adjusted to ensure the remaining balance of revenue - service costs (including tax) is zero or slightly positive, allowing the service to break even.

B. Results

1) Service Strategies: Our results show that the WeeDrive service can accommodate up to 260 students (642 trips per day) with no rejections, under both the flexible and moderate levels of strictness. However, the inflexible scenario can only achieve zero rejections with a demand of 130 students (317 trips per day). Figure 3 shows the percentage of rejected trips for different student population sizes and operational strictness scenarios, considering a single 45-minute break and staggered driver schedules. The main KPIs for these scenarios are also shown in Table IV. Passenger experiences in the moderate and flexible scenarios are quite similar, whereas in the inflexible scenario, waiting times—particularly at home locations—are significantly shorter. Additionally, maximum travel times are reduced by approximately 20 minutes compared to the other scenarios. The average daily cost per student ranges from €14 in the moderate and flexible scenarios to €27 in the inflexible one.

The impact of the different scheduling strategies is shown in Tables VI and VII. Implementing simultaneous start times results in a 15% to 23% decrease in the number of students served in both flexible and moderate scenarios. This decrease leads to an 18% to 30% increase in the cost per student. However, in the inflexible scenario, results were nearly identical for both start-time



(a) Inflexible

(b) Moderate

(c) Flexible

Fig. 3: Percentage of trips served and rejected by number of students and inflexibleness levels

TABLE III: Main KPIs notation

Abbreviation	Description
NoS_0rej	# of students served without rejections
NoT_0rej	# of trips served without rejections
AW_{sc,ac,ho} (mins)	Average waiting time in schools|activities|homes
TTPT_{avg,max} (mins)	Average|Maximum travel time / trip
ADPTS (km)	Average Distance per trip served
TDSC (€)	Total daily service cost
ADSC (€)	Average daily student cost

TABLE IV: Main KPIs for the scenarios with staggered start and a single 45-minutes break

Units	Flexible	Moderate	Inflexible
NoS_0rej	260	260	130
NoT_0rej	642	642	317
AW_{sc,ac,ho}	16|15|25	13|9|24	8|7|9
TTPT_{avg,max}	28|77	30|75	22|58
ADPTS	1.88	1.90	3.00
TDSC	3528	3530	3491
ADSC	13.6	13.6	26.9

strategies, indicating no negative effect. Regarding break schedules, splitting breaks leads to worse outcomes across all scenarios. This result is counterintuitive, as it should perform at least as well as the single-break scenarios. This suggests that the heuristic used for split-break scenarios is converging to a local optimum rather than a global one.

TABLE V: Performance Difference notation

Abbreviation	Description
ΔNoS_0rej (%)	Change in the # of students served without rejections
ΔADSC (%)	Change in cost per student

2) Pricing Policies: The best performing service strategy of moderate strictness level was selected to evaluate the 3 different pricing policies with the objective of comparing these policies from the user (parent/student) perspective, given the fact that this service is at an early implementation stage and user satisfaction is of the utmost importance for increasing the platform's user-base. As mentioned in Section III, the pricing policies are derived ex post the routes calculation, illustrated in Figure 4. The equations for the Taxi Fare and the 3 selected pricing policies can be found on Table VIII

For pricing policy Pol_1, the min, max and avg total fares per trip are €3.70, €10, and €6.54 respectively. For

TABLE VI: Impact of drivers' breaks distribution on key KPIs

	Simultaneous vs. staggered start		
	Flexible	Moderate	Inflexible
ΔNoS_0rej	-15	-23	0.0
ΔADSC	18	30	0

TABLE VII: Impact of drivers' breaks division on key KPIs

	Divided vs. single break		
	Flexible	Moderate	Inflexible
ΔNoS_0rej	-27	-50	-31
ΔADSC	40	29	30

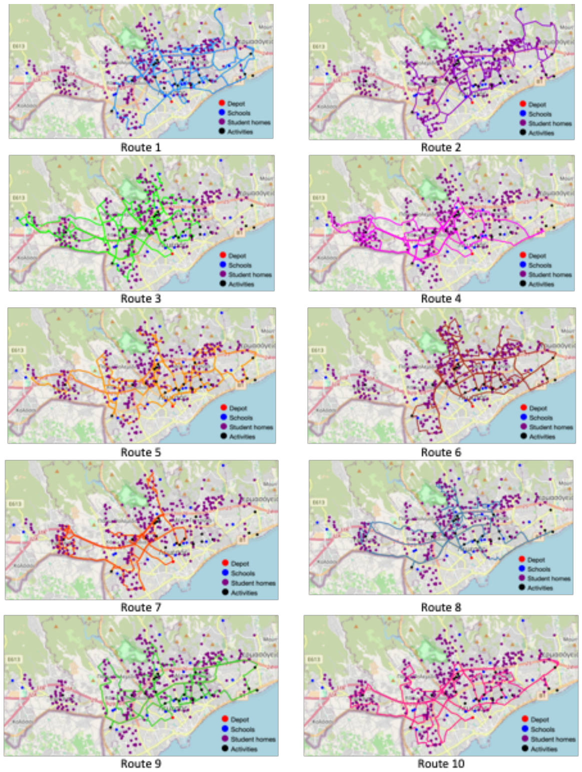


Fig. 4: Illustration of Routes

TABLE VIII: Pricing Policies

Name	**Total Fare (€)**
$TaxiF$	$3.42 + D0.73$
Pol_1	$min\{4.5 + D0.5, TaxiF\}$
Pol_2	$min\{6 + D0.3, TaxiF\}$
Pol_3	$min\{16.6/AOpv + D0.6, TaxiF\}$

pricing policy Pol_2, the min, max and avg total fares per trip are €3.70, €9.30, and €6.54 respectively. For the most equitable pricing policy, Pol_3, the min, max and avg total fares per trip are €3.11, €11.55, and €6.54 respectively. Assuming that the total fare per trip is normally distributed, Pol_3 has longer tails than the distributions of Pol_1 and Pol_2. The total fares paid by each student per trip are aggregated to derive the ECDF (Empirical Cumulative Distribution Function) for the total student fares per day, shown in Figure 5. As is evident from the ECDF, for policies Pol_1 and Pol_2, more than 50% of the student population is charged from €16-€24 per day. On the other hand, Pol_3 charges daily total student fares not exceeding €18 for 100% of the student population. From a user satisfaction standpoint, Pol_3 is the best choice.

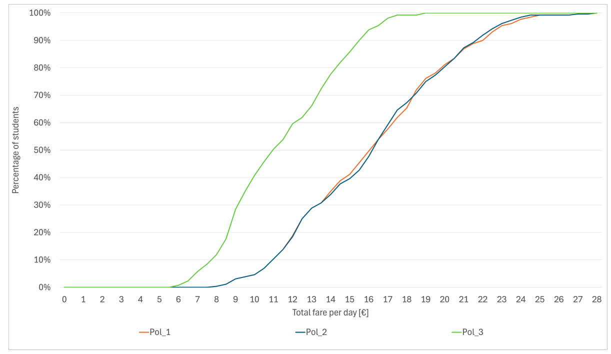


Fig. 5: ECDF for total student fare per day

IV. Conclusion

This study explores a microtransit service for transporting students to after-school activities, focusing on user experience and financial sustainability. The methodology combines a trip generation process to estimate demand with a Vehicle Routing Problem with Pickup and Delivery with Time Windows (VRPPDTW) to optimize daily routes. The methodology is applied to Limassol, Cyprus, considering various demand levels, service time window strictness, fare scenarios, and driver schedule strategies. The results show that allowing up to 30-minutes waiting time and 60-minutes detour can help to accommodate more students and to improve the financial sustainability (with a small surplus being achieved). Additionally, strategically scheduled driver shifts and a single 45-minute break lead to significantly better outcomes. Although splitting the break into two periods (15 minutes and 30 minutes) should further improve results, the heuristic tends to converge on a local optimum. We have also selected the moderate strictness level service strategy, as the best performing strategy, to implement 3 different pricing policies and compare them from the user perspective. We found that, assuming all policies being equal in helping us to break even, the policy that takes into account the partial trip overlap, Pol_3 is most likely to increase user satisfaction. Future research should focus on refining the heuristic for the separate break strategy and examining how demand responds to variations in fare and service characteristics.

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